CM617 · SUSTAINABILITY PRINCIPLES AND PRACTICES

TEAM EVISION

EV vs ICE: when does an electric vehicle repay its carbon debt?

Lifecycle break-even across vehicles, grids and countries, with a focus on India

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Research Questions

- How does grid carbon intensity affect per-kilometre and lifetime emissions savings for an EV versus a similar petrol vehicle?
- How does carbon break-even distance vary by country, vehicle class, annual mileage, lifetime, and EV energy consumption?
- Are EVs adopted fastest where modeled emissions savings are greatest after controlling for charging infrastructure, policy, and economic conditions?
- Why is EV adoption slow in India?

Scope and Method

The study compares three electric–petrol pairs of the same model across nine national grids, and computes the distance at which the EV's lower running emissions repay its battery's manufacturing emissions.

- Vehicles: a scooter, a hatchback and a mid-size SUV, each in petrol and battery-electric form, using Indian models with published real-world test data.
- Countries: India, Poland, China, Japan, the United States, Germany, the United Kingdom, France and Norway, chosen to span coal-heavy to near-zero-carbon grids.
- Boundary: well-to-wheel for petrol, lifecycle generation emissions plus grid and charging losses for electricity, and battery production for manufacturing. The rest of the vehicle is assumed to cost the same CO_2 to build in either version, so it cancels out.

VEHICLE PAIRS

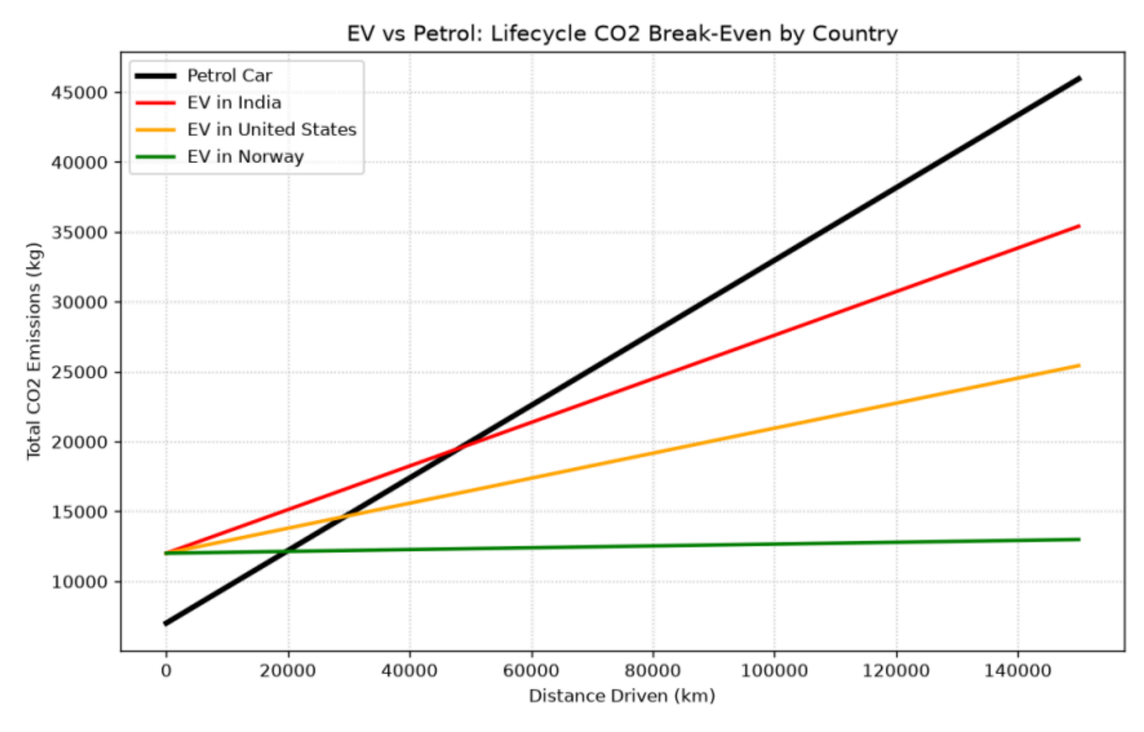
Real-world tests set the energy use of each EV and its petrol twin

Segment	EV (battery)	EV use at plug	Petrol twin (real-world)	Petrol g/km	Battery debt
Scooter	TVS iQube (3.5 kWh)	0.034 kWh/km	Activa 110 (50 km/L)	59	245 kg
Hatchback	Tata Tiago EV (24 kWh)	0.143 kWh/km	Tiago 1.2 (15 km/L)	196	1.7 t
Mid-size SUV	Tata Harrier EV (75 kWh)	0.208 kWh/km	Harrier 1.5T (9.33 km/L)	315	5.3 t

Official ARAI mileage overstates petrol efficiency and EV range, so measured figures are used. Battery debt = battery kWh × 70 kg CO_2e/kWh.

BASELINE · 150,000 KM

EV vs Petrol Lifecycle Break-Even by Country



~20k km

break-even in Norway

~29k km

break-even in the US

~48k km

break-even in India

Simple Takeaway

- Making EV batteries creates more CO_2 than making normal cars.
- EVs only cut emissions if they use clean power.
- In places like Norway (hydro power), EVs save a huge amount of CO_2.
- In places like India (coal power), EVs save much less CO_2.

BREAK-EVEN DISTANCE

Every EV breaks even in every country

Break-even (km)	Norway	France	US	India
Two-wheeler	6,273	6,427	8,676	13,938
Sedan	17,608	18,067	24,880	41,954
SUV	20,425	20,930	28,271	45,492

Clean grids

Net lifetime savings of 80–89% in Norway and France.

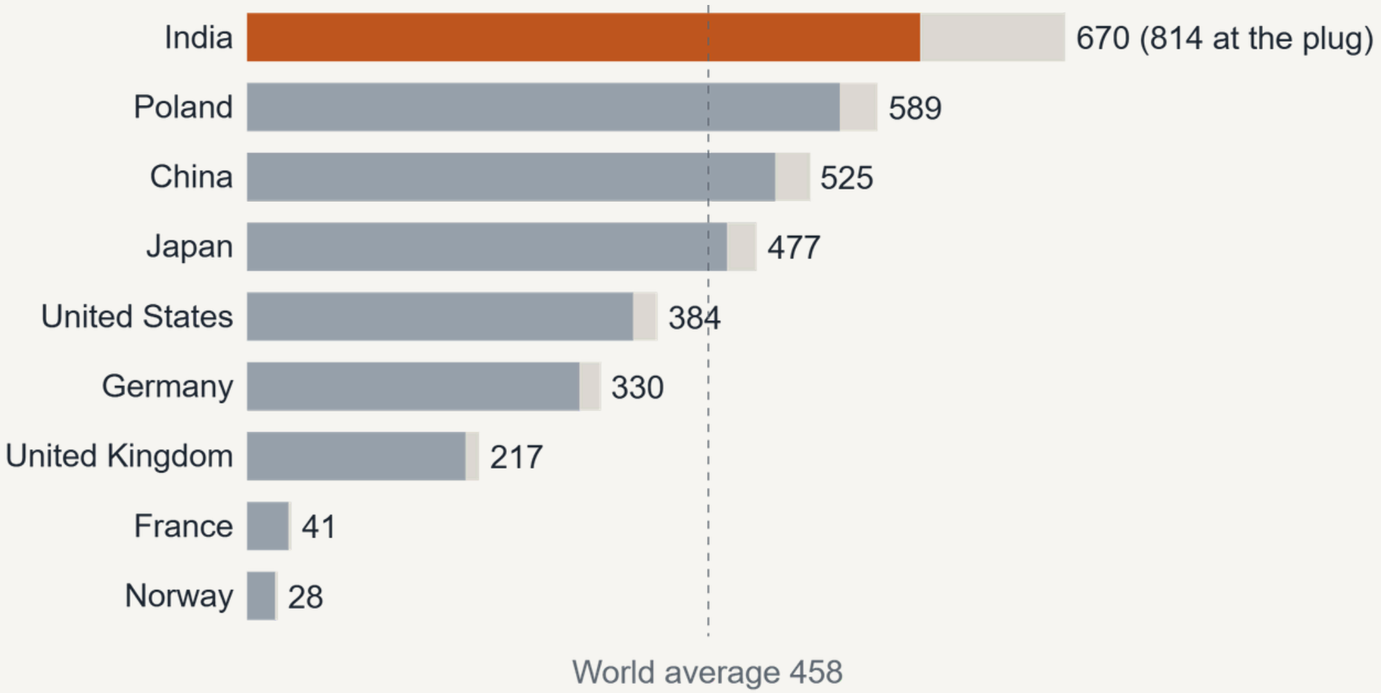
Coal-heavy grid

India still saves 30–37% and repays within 0.8–2.8 years.

Dataset used: fueleconomy.gov

GRID CARBON INTENSITY

India’s grid is 24 times as carbon-intensive as Norway’s



670

g CO_2e/kWh as generated in India, 814 at the plug

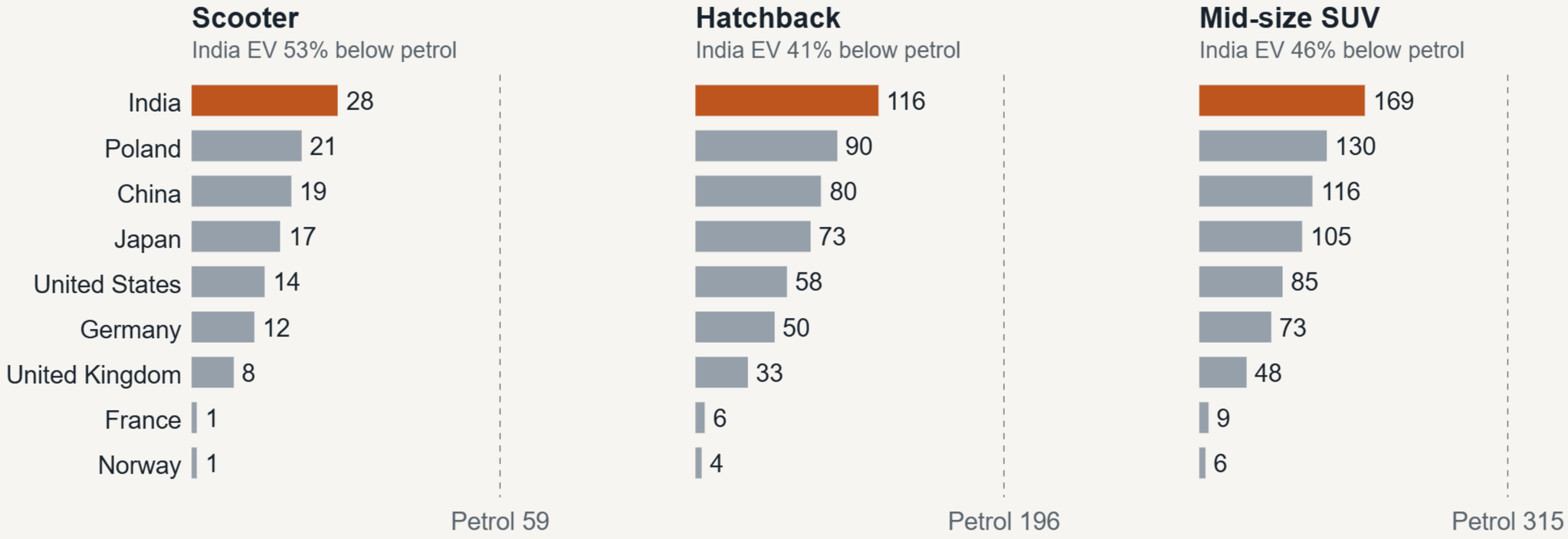
28

g CO_2e/kWh in Norway

Grey = as generated; pale = added by grid losses. India loses 17.7% in transmission, others 6%.

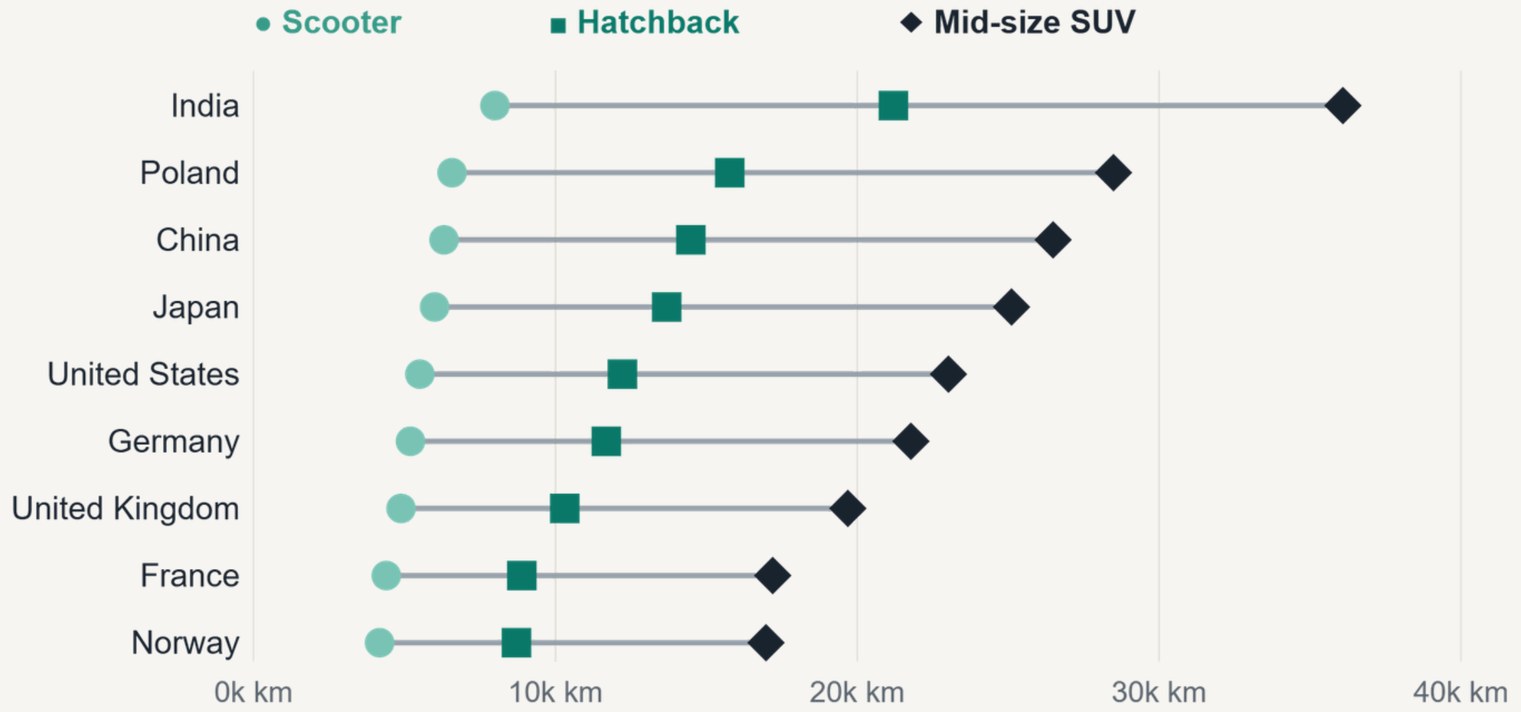
PER-KM EMISSIONS

Even on India's grid, each EV emits 41–53% less CO_2 per km than its petrol twin



BREAK-EVEN DISTANCE

From 4,200 km for a scooter in Norway to 36,100 km for an SUV in India



4,200 km

scooter in Norway, 0.5 yr

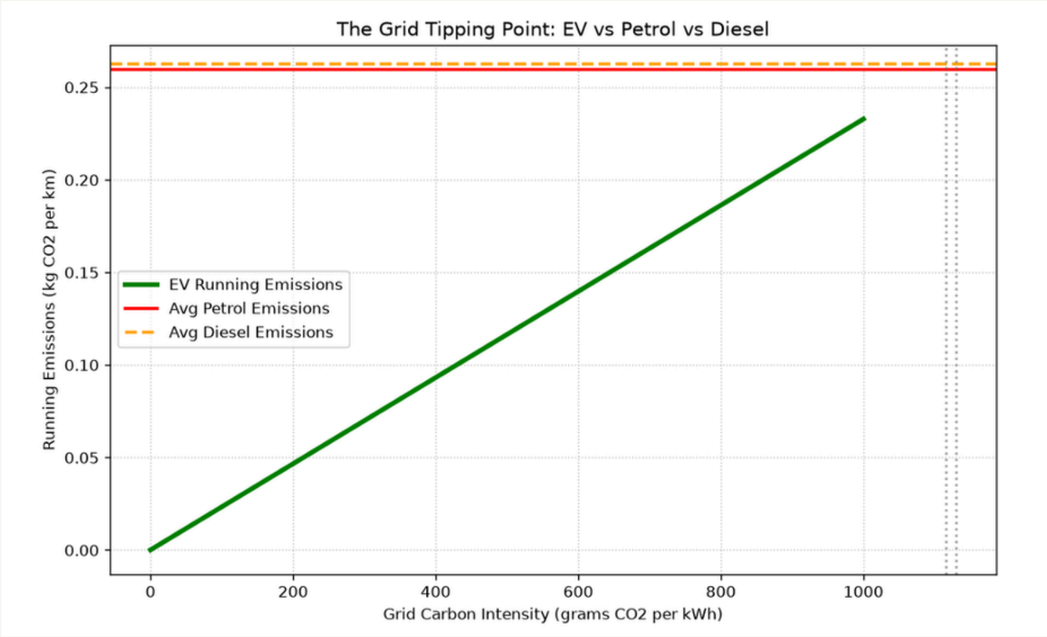
36,100 km

SUV in India, 3.0 yr

SUV needs 4–5× the scooter's distance in every country; India's grid adds only 1.9–2.4× vs Norway.

THE TIPPING POINT

A grid would have to be dirtier than all-coal to beat petrol



Grid group	Per-km cut vs petrol
France, Norway	97–98%
UK, Germany	74–87%
US, Japan	63–76%
China, Poland	54–68%
India	41–53%

Grids run from 28 g/kWh in Norway to 670 in India.

Tipping point, petrol

1,115

g CO_2/kWh; diesel 1,130

By vehicle type

1,300–1,600

g CO_2/kWh. Real lifecycle grids peak near 670–1,000.

WHERE EVs SUIT BEST

The cut depends on the grid; only an impossibly dirty grid erases it

Grid group	Per-km cut	Verdict
France, Norway (near zero)	97–98%	Clear win; under 1.5 years even for the SUV
UK, Germany (low)	74–87%	Strong win; under 2 years for all
US, Japan (mid)	63–76%	Strong win; US drivers break even faster
China, Poland (coal-heavy)	54–68%	Solid win; 0.7–2.4 years
India (coal plus high losses)	41–53%	Win for scooters and hatchbacks; SUV depends on petrol use

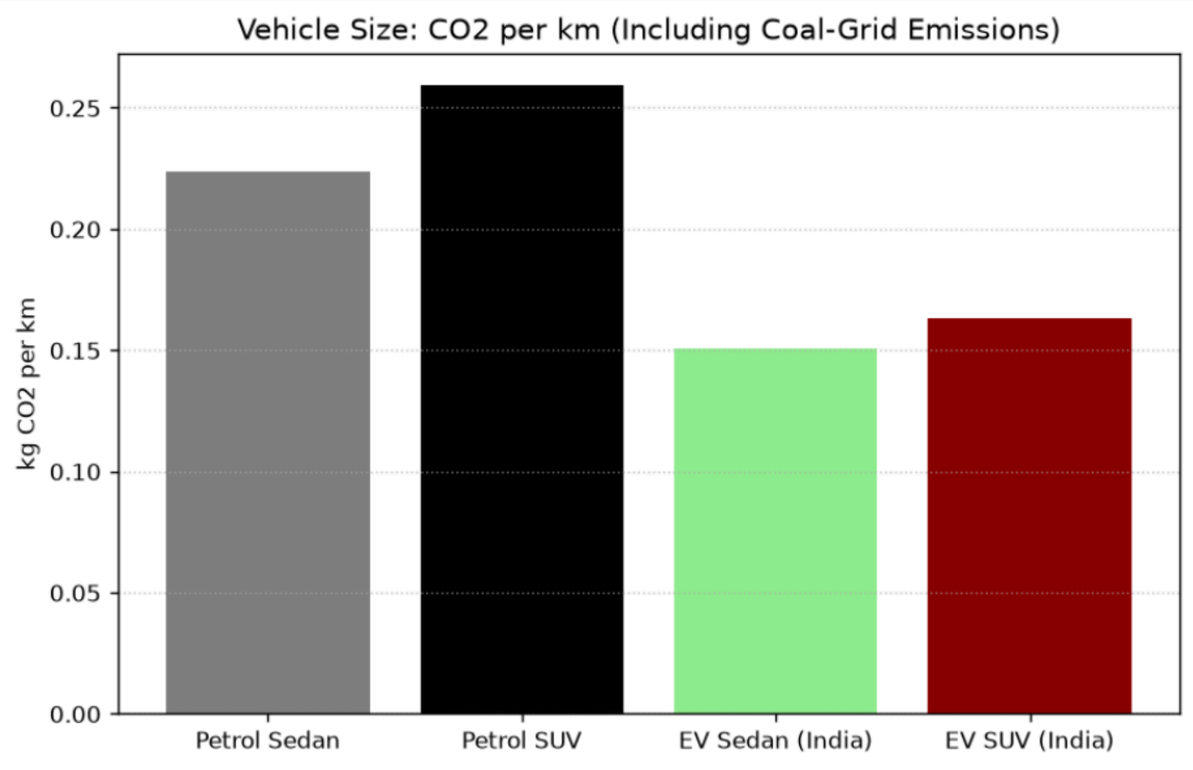
1,286
g/kWh tipping point, hatchback

1,423
g/kWh tipping point, SUV

1,631
g/kWh tipping point, scooter

800–1,000
g/kWh for an all-coal grid

The Impact of Vehicle Size (SUV vs Sedan)



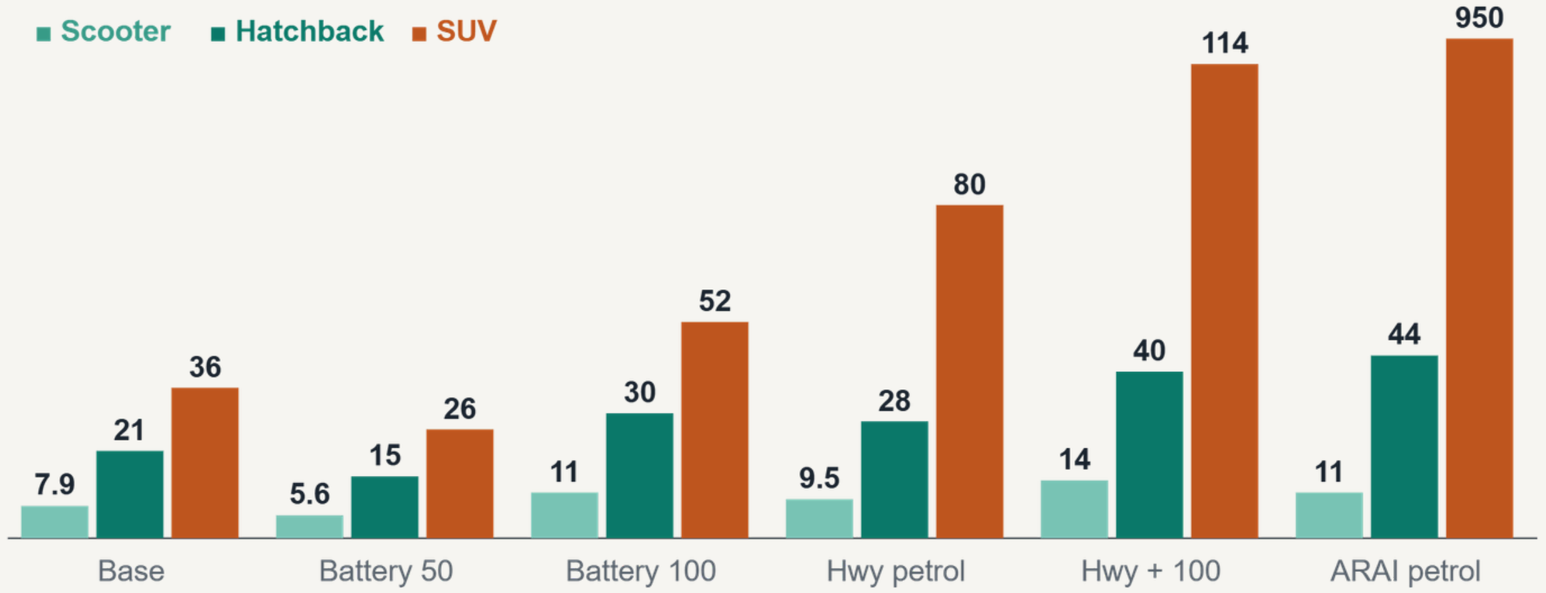
kg CO_2 per km, India grid

Simple Takeaway

- Petrol cars emit the highest CO_2 (SUVs: 0.259 kg/km, Sedans: 0.224 kg/km).
- EVs emit less CO_2 per km, even in India (SUVs: 0.163 kg/km, Sedans: 0.150 kg/km).
- Large EV SUVs pollute almost as much as small petrol sedans because they are heavy.
- Small EV sedans are much better for the environment than big EV SUVs.
- Government subsidies should focus on small EVs instead of big luxury EV SUVs.

SENSITIVITY · INDIA

Only the large SUV can fail to break even in the nine-grid study



950k km

SUV if petrol gets its ARAI mileage

At ARAI mileage the petrol Harrier emits about 175 g/km vs 169 for the EV, a 6 g/km gap, so it effectively never catches up. ARAI is a lab upper bound. Bars capped at 120k km.

INDIA · COMPARISON OF SCOOTER, SUV AND HEAVY VEHICLE

Scooters pay back fastest; buses have the thinnest margin

	Nexon EV vs petrol	Ola S1 vs Activa	E-bus(Switch EiV 12) vs diesel bus
Break-even	≈36,000 km	≈6,900 km	≈175,000 km
Share of vehicle life	17.9%	5.7%	20.3%
Per-km saving	82 g	30 g	110 g
Lifetime CO_2 saving	32%	48%	8%

Grid 670 g/kWh, battery 68 kg/kWh, real-world efficiency, well-to-wheel fuel emissions.

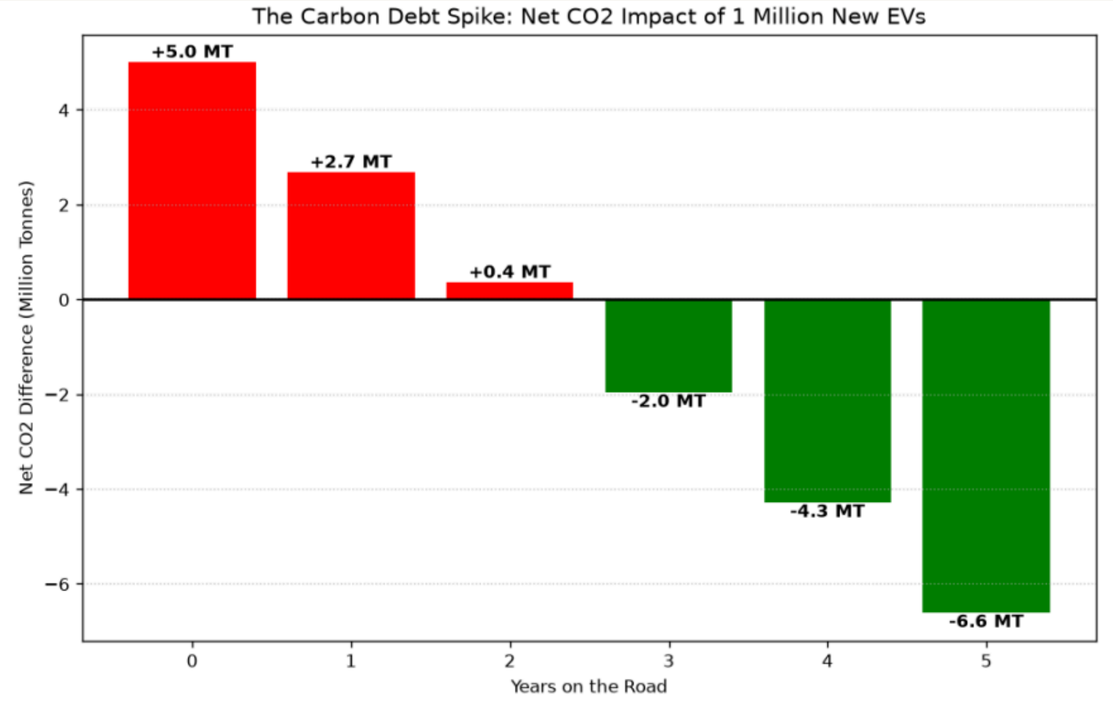
STRESS TESTS

The results are robust for small EVs and get fragile for large ones

Vehicle	Base case	Stress test	Break-even	Takeaway
Scooter	6900 km	Worst case (battery 196 kg/kWh)	19,800 km, inside life	Break-even in every tested scenario; worst case still within 120,000 km assumed life.
Nexon EV	36000 km	Petrol at ARAI mileage	95,600 km (+165%)	Usually breaks even, but high-carbon grid or optimistic petrol mileage can push it beyond vehicle life
Nexon EV		Grid at 1,100 g/kWh	259,000 km, beyond 188,000 km life	
Electric bus	175000 km	1.3 kWh/km, diesel 3.5 km/L or 1,100 g/kWh grid	Never	Most fragile result; outcome depends strongly on grid intensity and real-world energy/fuel efficiency.

CARBON DEBT

The Carbon Debt Spike (Short-Term Impact)



Simple Takeaway

- Building 1 million new EVs causes an immediate spike in pollution in Year 0.
- Large battery factories need massive energy before a single car is sold.
- Driving an EV generates less daily pollution than driving a petrol car.
- It takes about 3 years of driving to pay off this initial carbon debt.
- True net climate benefits only start after passing this 3-year mark.

ADOPTION

India’s EV market follows the best carbon case, but cars lag

2.1M

two- and three-wheeler EV sales, 10% share

170k

EV car sales, 4% share

+53%

average BEV price premium (~$23k vs $15k)

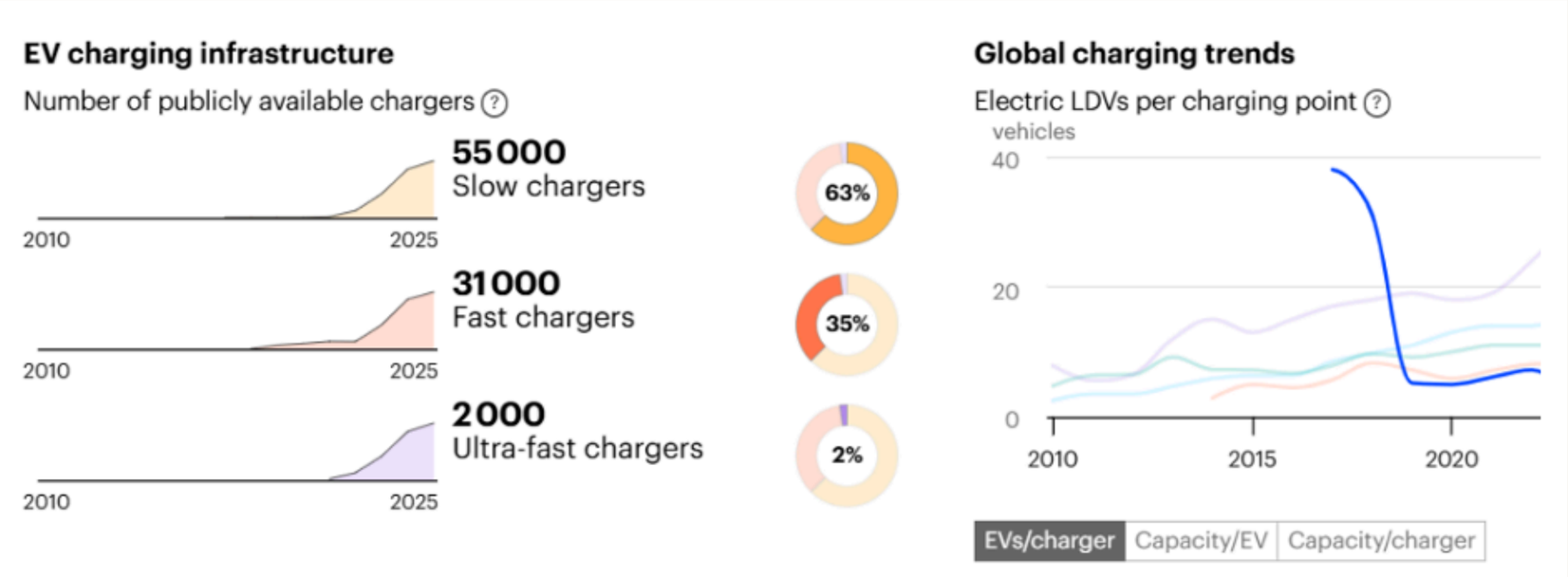
24%

renewable electricity in India vs 59% in Germany

IEA projects Indian two- and three-wheeler stock rising from 8.7M to 44M by 2035, against 0.41M to 3.5M cars. Price gaps are small for city cars (+12%) but large for medium cars (+189%).

ADOPTION

Why is EV adoption slow in India?



- Price: the average electric car costs about 53% more than a comparable petrol car
- Charging: India has about 55,000 slow, 31,000 fast and 2,000 ultra-fast public chargers (63% / 35% / 2%).
- Chargers per EV: EVs per charger fell from about 38 in 2017 to a low single-digit figure by 2025. Charger numbers are not the main bottleneck today, but ultra-fast capacity is thin.
- Income and cities: India has 10.2 EVs per 1,000 people, and only 35.7% of Indians live in cities.

Conclusions

EVs win everywhere

All 27 vehicle-country cases break even within about 3 years.

Grid is the lever

India takes 1.9–2.4× longer than Norway; clean power widens the gain.

Prioritise small EVs

Scooters and small cars give the best return; buses and large SUVs are fragile in India.

Adoption

Adoption is highest for two-wheelers, where savings are best, and lowest for cars, where price is the barrier.

Caveats: static grids, battery-only manufacturing in some reports, US-based efficiency data, carbon only (no cost or end-of-life). Figures differ across reports because of different assumptions.

Work Distribution

Members	**Work Done**
Anjali	Break-even of 9 countries for 3 vehicle types (fueleconomy.gov dataset), and EV adoption in India
Shreya	Break-even of 9 countries for 3 vehicle types (Our World in Data), and petrol vs EV comparison of SUV, hatchback and two-wheeler
Manasvi	Indian vehicle comparison (Nexon, scooter, bus) and stress test (sensitivity analysis)
Riddhi	Grid tipping point, and petrol vs EV comparison of SUV and sedan